

PRUDHOE BAY DEADHORSE, AK, USA

WMO: 700637

Lat: 70.192N

Lon: 148.477W

Elev: 61

StdP: 14.66

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 27406

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.1	(c) -39.9	(d) -35.9	(e) 0.7	(f) -28.3	(g) -34.5	(h) 0.8	(i) -28.1	(j) 39.5	(k) -5.8	(l) 35.4	(m) -4.0	(n) 7.4	(o) 240	(p) 1.068

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	67.4	57.3	63.0	55.1	58.7	52.9	58.2	65.5	55.6	61.9	53.2	58.5	11.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	62.7	61.2	51.8	57.3	59.3	49.5	52.5	57.1	25.1	65.6	23.6	62.3	22.1	58.7	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 33.7	(b) 29.2	(c) 26.4	(e) -45.8	(f) 76.7	(g) 4.5	(h) 3.6	(i) -49.0	(j) 79.3	(k) -51.7	(l) 81.5	(m) -54.2	(n) 83.5	(o) -57.5	(p) 86.2		
(a) 33.7	(b) 29.2	(c) 26.4	(e) -35.1	(f) 62.2	(g) 3.0	(h) 2.8	(i) -37.2	(j) 64.2	(k) -38.9	(l) 65.8	(m) -40.6	(n) 67.4	(o) -42.7	(p) 69.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	-14.1	-13.3	-13.3	3.5	24.0	40.1	47.6	43.9	35.1	18.8	2.4	-8.4
	DBStd	25.47	14.00	15.06	14.20	11.99	9.61	7.81	7.02	7.38	6.70	10.02	11.93	13.14
	HDD50	13247	1987	1773	1962	1395	807	317	132	223	447	966	1427	1811
	HDD65	18610	2452	2193	2427	1845	1272	748	539	654	896	1431	1877	2276
	CDD50	113	0	0	0	0	0	20	58	34	1	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	17	0	0	0	0	0	2	12	3	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	12.3	12.9	13.0	12.6	12.2	12.9	12.8	11.6	11.1	11.5	12.2	12.4	12.4
Precipitation	PrecAvg	7.3	0.4	0.3	0.3	0.2	0.3	0.6	1.1	1.5	1.1	0.9	0.5	0.4
	PrecMax	13.2	1.2	0.9	1.3	0.5	0.9	1.7	2.6	3.5	3.0	2.0	1.3	1.3
	PrecMin	4.3	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.4	0.2	0.4	0.1	0.0
	PrecStd	1.9	0.3	0.2	0.2	0.1	0.2	0.4	0.5	0.8	0.7	0.4	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	28.1	28.2	35.9	47.3	70.4	75.3	71.2	61.0	39.3	32.0	28.4
		MCWB	29.5	26.7	26.4	33.5	43.3	56.7	60.9	59.6	51.6	37.1	31.3	26.9
	2%	DB	20.6	20.6	20.6	30.1	40.6	63.9	68.8	64.9	53.1	34.7	26.8	20.2
		MCWB	19.7	19.7	19.6	28.8	37.7	54.4	58.4	56.4	47.4	33.4	25.6	19.4
	5%	DB	10.3	14.6	13.2	25.4	37.2	58.6	64.0	60.5	48.3	32.5	22.9	14.4
		MCWB	9.4	13.5	12.2	24.0	35.2	51.8	56.2	54.3	44.9	31.5	21.9	13.5
	10%	DB	3.7	7.4	5.7	20.8	35.0	53.2	59.6	55.8	44.2	30.6	19.3	9.2
		MCWB	3.1	6.4	4.7	20.0	33.4	48.5	53.9	51.7	42.2	29.8	18.8	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.3	27.5	27.0	33.8	43.4	57.9	62.1	60.9	51.7	37.6	31.4	27.8
		MCDB	31.7	28.3	28.3	35.7	47.0	67.8	72.6	69.6	59.3	38.8	31.9	28.8
	2%	WB	20.9	20.6	20.5	28.9	37.8	54.8	59.1	57.4	48.2	33.6	25.6	19.9
		MCDB	21.8	21.6	21.3	30.0	40.3	63.0	67.2	62.9	52.6	34.4	26.7	20.5
	5%	WB	11.5	14.8	13.8	24.3	35.3	51.8	56.6	54.7	45.3	31.7	22.0	14.2
		MCDB	12.4	15.8	14.7	25.5	37.0	58.3	63.3	59.8	48.3	32.4	22.8	14.9
	10%	WB	4.5	8.2	6.3	20.0	33.4	48.3	54.3	51.7	42.1	29.9	18.8	8.9
		MCDB	5.2	9.2	7.3	20.7	34.6	53.3	59.7	55.5	44.2	30.6	19.4	9.6
Mean Daily Temperature Range	5% DB	MDBR	12.3	12.7	12.8	13.8	9.0	11.7	14.1	11.3	8.8	9.6	12.0	12.0
		MCDBR	21.4	19.8	16.5	15.4	11.1	22.7	22.8	20.6	18.0	8.6	12.9	17.2
		MCWBR	20.8	19.4	16.2	14.6	9.0	14.1	13.5	12.3	12.4	7.9	12.7	16.8
	5% WB	MCDBR	21.9	19.9	16.8	15.6	10.3	22.2	21.8	19.3	17.0	8.4	13.0	17.3
		MCWBR	21.3	19.4	16.5	14.8	8.4	14.2	13.8	12.2	12.0	7.8	12.8	16.9
Clear-Sky Solar Irradiance	taub		N/A	0.202	0.242	0.298	0.319	0.335	0.347	0.329	0.295	0.218	N/A	N/A
	taud		N/A	2.063	2.220	2.134	2.205	2.350	2.423	2.515	2.525	2.167	N/A	N/A
	Ebn at Noon		0	173	249	263	272	271	263	256	233	161	0	0
	Edh at Noon		0	12	22	33	36	32	29	23	17	10	0	0
All-Sky Solar Radiation	RadAvg		0	115	625	1373	1885	1961	1616	951	484	148	0	0
	RadStd		0	8	26	61	103	135	126	99	45	19	0	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.05	N/A	N/A	N/A	N/A	N/A	-902	-889	N/A	N/A
(47) Regional (3 neighbors)		+2.05	N/A	N/A	N/A	N/A	N/A	-902	-889	N/A	N/A

Nomenclature: See separate page